

Pension Pillars under Informality:

General-Equilibrium Effects of Reallocating Contributions

Oliver Pardo*

July 2026

Abstract

This paper studies how pension architecture affects labor informality, capital accumulation, and fiscal adjustment. I build a two-period overlapping-generations model with heterogeneous productivity, endogenous formal–informal choice, present bias, funded accounts, a pay-as-you-go (PAYG) program, and an endogenous consumption tax. The exercise compares a dual-assignment benchmark—lower-productivity formal workers in funded accounts and higher-productivity formal workers in PAYG—with pillars: every formal worker contributes to PAYG below a ceiling and to a funded account above it. When PAYG receipts are approximately matched, pillars raise informality by 6.2 percentage points, lower capital by 5.4 percent, and require a 5.3 percentage-point consumption-tax increase. Old-age consumption falls by 10.2 percent and experienced welfare by 1.0 percent in consumption-equivalent terms. An expanded public pillar produces larger effects but raises PAYG receipts by 79 percent, so it is not an apples-to-apples comparison. Contribution incidence can matter as much as the public program’s aggregate size.

Keywords: pensions; informality; overlapping generations; PAYG; funded accounts; present bias. **JEL:** E21; H55; J26; J46.

1 Introduction

Pension reform in an economy with a large informal sector has a general-equilibrium trade-off. Mandatory contributions can protect present-biased workers from insufficient

*Departamento de Administraci3n, Pontificia Universidad Javeriana, Bogot3, Colombia. Email: pardoo@javeriana.edu.co.

retirement saving, but they also change the return to formal work. When workers can move across sectors, the distribution of contributions affects the tax base, aggregate saving, and fiscal closure.

This paper asks what happens when a pension system moves from separate funded and PAYG assignments to a pillar architecture. In the benchmark, lower-productivity formal workers are assigned to funded accounts and higher-productivity formal workers to PAYG. Under pillars, the first tranche of every formal worker's earnings finances PAYG and earnings above a ceiling finance an individual account. This is a comparison between *architectures*, not a model in which workers choose between pension regimes. The individual choice is formal versus informal employment.

The distinction drives the result. Funded contributions earn the market return; PAYG benefits follow a replacement rule and the growth of the contribution base. Extending the PAYG tranche to workers near the sectoral margin reduces their value of formality. Some become informal, funded saving and capital fall, and a narrower tax base requires a higher consumption tax.

The main exercise approximately holds PAYG scale fixed: receipts are within 1.2 percent of the benchmark. Informality rises from 56.4 to 62.5 percent, capital falls 5.4 percent, and the consumption tax rises from 17.0 to 22.3 percent. An expanded-pillar exercise yields larger effects but also raises PAYG receipts 79 percent; it therefore combines architecture and scale.

The contribution is threefold. First, the exercise isolates contribution incidence across heterogeneous workers. Second, it integrates informality, present-biased saving, funded and PAYG pensions, and fiscal closure. Third, it separates decision utility from experienced welfare. The appendix gives the numerical algorithm.

2 Related literature

The framework follows the OLG approach to social security in [Diamond \(1977\)](#). Mandatory retirement programs can address undersaving ([Feldstein, 1985](#)); present-biased preferences provide a behavioral rationale ([Laibson, 1998](#); [İmrohoroglu et al., 2003](#)). [Pardo \(2023\)](#) studies mandatory saving with an informal labor market. Here the question is how funded and PAYG contributions are allocated across formal workers.

Social insurance cannot be evaluated independently of informality. Payroll-financed benefits alter formal employment incentives ([Levy, 2008](#)), while informality reflects worker and firm choices ([Ulyssea, 2018](#)). Search and human-capital accumulation shape

sector transitions (Bobba et al., 2021); the shadow economy changes optimal redistribution (Doligalski and Rojas, 2023). Pension models with informality include Joubert (2015) and McKiernan (2020). Related work studies retirement and occupational behavior when formal contributions can be avoided (Ferreira and Parente, 2020; Moreno, 2022). This paper focuses on which workers bear the low-return pension tranche and how close they are to the informality margin.

3 Model

3.1 Demography, types, and preferences

One period is approximately 40 years. Population grows at rate n , productivity at g , and an agent survives to old age with probability $1 - m$. Type $i \in [0, 1]$ has density

$$f(i) = \frac{2.4450}{1.006341403957648} \exp \left[- \left(\frac{i - 0.4655}{0.2329} \right)^2 \right], \quad \int_0^1 f(i) di = 1. \quad (1)$$

Sector-specific efficiency is

$$z_f(i) = \text{base}_f e^{\rho_f i}, \quad z_s(i) = \text{base}_s e^{\rho_s i}. \quad (2)$$

With $u(c) = (c^{1-\sigma} - 1)/(1 - \sigma)$, decision and experienced utilities are

$$V(i) = u(c_1(i)) + \beta(1 - m)u(c_2(i)), \quad W(i) = u(c_1(i)) + (1 - m)u(c_2(i)). \quad (3)$$

Thus $\beta < 1$ distorts behavior but is not a normative welfare weight.

3.2 Technology, households, and sector choice

Formal and informal output are

$$Y_f = A_f K^\alpha N_f^{1-\alpha}, \quad Y_s = A_s N_s.$$

Competitive pre-tax prices satisfy

$$\tilde{r} + \delta = \alpha A_f (N_f/K)^{1-\alpha}, \quad \tilde{w}_f = (1 - \alpha) A_f (K/N_f)^\alpha, \quad \tilde{w}_s = A_s.$$

Taxes map these into the net return r and effective wages w_f, w_s .

For sector $j \in \{f, s\}$ and voluntary saving $a_j(i) \geq 0$,

$$(1 + \tau_c)c_{1j}(i) + a_j(i) = y_{1j}(i), \quad (4)$$

$$c_{2j}(i) = \frac{1+r}{1+g}a_j(i) + p_j(i). \quad (5)$$

Formal income is $y_{1f} = (1 - \tau_\ell - \tau_p)w_f z_f(i) + T_{1f}$ and informal income is $y_{1s} = w_s z_s(i) + T_{1s}$. Baseline transfers are zero. After solving saving in each sector,

$$d_f(i) = \mathbf{1}\{V_f(i) \geq V_s(i)\}.$$

The formal set is computed point by point rather than imposed as an interval.

3.3 Pension architectures

In the *dual-assignment benchmark*, $D(i) = \mathbf{1}\{i \geq \bar{i}\}$ and

$$p_f^D(i) = [1 - D(i)] \frac{(1+r)\tau_p w_f z_f(i)}{1+g} + D(i) \frac{(1-m)b w_f z_f(i)}{1+g}. \quad (6)$$

Lower types are funded and higher types PAYG. Assignment is exogenous; workers do not choose their pension regime.

Under *pillars*, every formal worker's first earnings tranche is PAYG and the excess is funded:

$$p_f^P(i) = \frac{(1-m)b \min\{w_f z_f(i), \bar{y}\}}{1+g} + \frac{(1+r)\tau_p \max\{w_f z_f(i) - \bar{y}, 0\}}{1+g}. \quad (7)$$

Pillars therefore shift PAYG incidence from high types to the lower earnings of all formal workers.

3.4 Aggregation and fiscal closure

$$N_f = \int d_f(i) z_f(i) f(i) di, \quad N_s = \int [1 - d_f(i)] z_s(i) f(i) di, \quad (8)$$

$$I = \int [1 - d_f(i)] f(i) di, \quad K' = \frac{\int [a_j(i) + a^p(i)] f(i) di}{(1+n)(1+g)}. \quad (9)$$

Here a^p is funded pension saving. With PAYG receipts R^{pg} and benefits B^{pg} ,

$$\tau_c \mathcal{C} + \tau_\ell w_f N_f + \tau_n w_f N_f + \tau_k \tilde{r} K + R^{pg} = G + \mathcal{T} + B^{pg}.$$

The consumption tax closes the budget. Equilibrium combines household optimality, sector choice, competitive prices, market clearing, the capital fixed point, and public balance.

4 Calibration

Annual rates are compounded over 40 years: $n = 1.013^{40} - 1$, $g = 1.004^{40} - 1$, and $\beta = 0.97^{40}$. Table 1 gives the remaining parameters.

Table 1: Baseline parameters

Parameter	Value	Meaning	Parameter	Value	Meaning
m	.4000	mortality	σ	2.5000	curvature
n	.6764	population growth	β	.2957	present bias
δ	1.0000	depreciation	g	.1731	productivity growth
α	.4000	capital share	A_s	.1000	informal scale
ρ_s	2.9000	informal gradient	ρ_f	4.6400	formal gradient
τ_ℓ	.1250	labor tax	τ_n	.1250	payroll wedge
τ_k	.3200	capital tax	τ_p	.1250	pension rate
b	.6400	PAYG replacement	G	.3300	government purchases
$\bar{i}$	.7358	assignment cutoff	$\bar{y}$	.3360	pillar ceiling

Steady states use 501 types. Capital and the consumption tax are joint fixed points. The main ceiling yields PAYG receipts 1.2 percent below the benchmark. The $\bar{y} = 1$ robustness is both an architecture change and a public-pillar expansion.

5 Results

5.1 Comparable PAYG scale

Pillars raise informality by 6.2 percentage points. Formal labor falls 7.7 percent and funded assets 8.7 percent, so capital declines 5.4 percent despite a small voluntary-saving increase. The smaller formal sector erodes tax bases and the consumption tax rises 5.3 percentage points. Young consumption falls 1.5 percent and old-age consumption 10.2 percent.

Table 2: Steady states with approximately matched PAYG receipts

Outcome	Dual assignment	Pillars	Change
Capital stock	.03429	.03245	−5.36%
Informal worker share	.56358	.62524	+6.17 pp
Effective formal labor	8.38358	7.74061	−7.67%
Effective informal labor	1.62865	1.89552	+16.39%
Consumption-tax rate	.16980	.22255	+5.28 pp
Annualized net return	.05456	.05421	−0.03 pp
Formal wage index	.05911	.05970	+0.99%
Young consumption	.46230	.45522	−1.53%
Old-age consumption	.26664	.23943	−10.21%
Voluntary capital	.01089	.01108	+1.76%
Funded pension capital	.02340	.02137	−8.68%
PAYG receipts	.01593	.01574	−1.19%
PAYG benefits	.02488	.02459	−1.19%

Annual return: $(1 + r)^{1/40} - 1$. pp denotes percentage points. Maximum equilibrium residuals are below 10^{-7} .

The mechanism is incidence. At the calibrated equilibrium, the funded tranche has a higher implicit return than PAYG. Dual assignment places PAYG on high types with a strong formal comparative advantage. Pillars extend it to workers near the sector margin, reducing $V_f(i) - V_s(i)$. Exit to informality reduces funded saving, narrows the formal tax base, and raises the consumption tax, reinforcing the response.

Decision utility falls from -4.1412 to -4.1834 and experienced welfare from -4.5563 to -4.6276 . By CRRA homogeneity, pillar consumption would need a uniform 1.04 percent increase to match benchmark experienced welfare. Decision-utility compensation is 0.68 percent.

5.2 Expanded public pillar

With $\bar{y} = 1$, informality reaches 77.1 percent and the consumption tax 45.2 percent. Because PAYG receipts rise 79 percent, these magnitudes quantify architecture plus scale.

5.3 Diagnostic transition

Capital is predetermined initially; informality and the tax respond immediately, after which the economy approaches the pillar steady state. This 101-type short path is a

Table 3: Robustness: expanded public pillar

Outcome	Dual	Expanded pillars	Change
Capital	.03429	.03039	−11.38%
Informal share	.56358	.77138	+20.78 pp
Consumption tax	.16980	.45209	+28.23 pp
Young consumption	.46230	.43033	−6.92%
Old-age consumption	.26664	.18519	−30.55%
PAYG receipts	.01593	.02858	+79.39%

Table 4: Selected dates in the transition to pillars

Date	Capital	Informal share	Consumption tax	Period return
1	.034153	.567143	.173010	7.369671
2	.034153	.577030	.174042	7.317255
3	.031988	.649558	.255602	7.167026
5	.032232	.635181	.235394	7.228585
10	.032461	.624755	.222091	7.263177
11	.032449	.625217	.222615	7.262405

numerical diagnostic, not a forecast.

6 Conclusion

Pension architecture changes the formal-employment margin even with an approximately fixed PAYG envelope. Moving to pillars shifts the PAYG tranche toward lower-earning formal workers, reduces funded saving and capital, narrows the formal tax base, and forces a higher consumption tax. In the comparable-scale experiment, informality rises 6.2 percentage points and experienced welfare falls about 1 percent in consumption-equivalent terms.

These results identify a mechanism, not a universal empirical claim. Pension assignment is exogenous, employment histories are absent, and the transition is diagnostic. A policy evaluation should estimate sector transitions, contribution density, benefit eligibility, administrative costs, heterogeneous mortality, and actual legal budget rules.

A Numerical algorithm

A.1 Stationary equilibrium

1. Build and normalize the type grid and quadrature weights.
2. Guess capital K and consumption tax τ_c and compute factor prices.
3. For every type and sector, solve the constrained one-dimensional saving problem and calculate benefits under (6) or (7).
4. Compare decision utilities point by point and aggregate labor, informality, consumption, saving, and PAYG flows.
5. Update capital from aggregate saving and τ_c from the public-budget residual, using damping when necessary. Iterate to tolerance.

The reported steady states use $Q = 501$ and maximum residual below 10^{-7} . A single formal cutoff is not imposed because pension kinks can create multiple crossings.

A.2 Matched PAYG and transition

For every candidate ceiling, the complete pillar equilibrium is recomputed. The outer objective is $h(\bar{y}) = R^{pg,P}(\bar{y}) - R^{pg,D}$. Grid kinks imply an approximate match: $\bar{y} = .336$ is within 1.2 percent.

For the transition, guess paths $\{K_t, \tau_{c,t}\}$, solve household and sector choices at each date, construct implied capital and tax paths, and update with damping. The diagnostic uses $Q = 101$, ten periods, and 334 iterations; its terminal capital gap is 2.1×10^{-4} . Checks cover positive consumption, nonnegative saving, density normalization, capital consistency, and government balance.

References

- Bobba, M., L. Flabbi, S. Levy, and M. Tejada (2021). Labor market search, informality, and on-the-job human capital accumulation. *Journal of Econometrics* 223(2), 433–453.
- Diamond, P. A. (1977). A framework for social security analysis. *Journal of Public Economics* 8(3), 275–298.

- Doligalski, P. and L. E. Rojas (2023). Optimal redistribution with a shadow economy. *Theoretical Economics* 18, 749–791.
- Feldstein, M. (1985). The optimal level of social security benefits. *The Quarterly Journal of Economics* 100(2), 303–320.
- Ferreira, P. C. and R. M. Parente (2020). Social security reform, retirement and occupational behavior. Working paper.
- İmrohoroglu, A., S. İmrohoroglu, and D. H. Joines (2003). Time-inconsistent preferences and social security. *The Quarterly Journal of Economics* 118(2), 745–784.
- Joubert, C. (2015). Pension design with a large informal labor market: Evidence from Chile. *International Economic Review* 56(2), 673–694.
- Laibson, D. (1998). Life-cycle consumption and hyperbolic discount functions. *European Economic Review* 42(3–5), 861–871.
- Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Brookings Institution Press.
- McKiernan, K. (2020). Social security reform in the presence of informality. Manuscript, Review of Economic Dynamics.
- Moreno, C. (2022). The impact of pension systems in labor markets with informality. Working paper.
- Pardo, O. (2023). Mandatory retirement savings in the presence of an informal labor market. *Journal of Population Economics* 36, 2857–2888.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review* 108(8), 2015–2047.